Worksheet 10.4

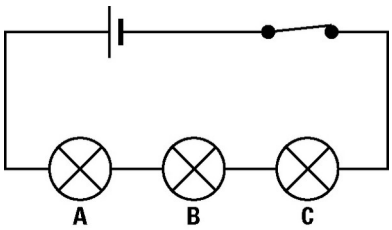
Science Quest 9: pages 358–60

Series and parallel circuits

Student: ..... Class: .....

Circuit one

1. Consider the following circuit containing identical lamps A, B and C in series.



(a) Compare the brightness of the three lamps.

.....

(b) Compare the currents that flow through lamp A and lamp C.

.....

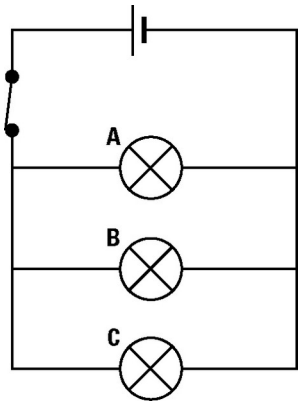
(c) If the filament in bulb B burns out, will any of the lights remain lit? Explain.

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.....

Circuit two

2. Consider the following circuit containing identical lamps A, B and C parallel. These lamps are the same as those for Circuit 1.



(a) Compare the brightness of the three lamps.

.....

(b) Compare the currents that flow through lamp A and lamp C.

.....

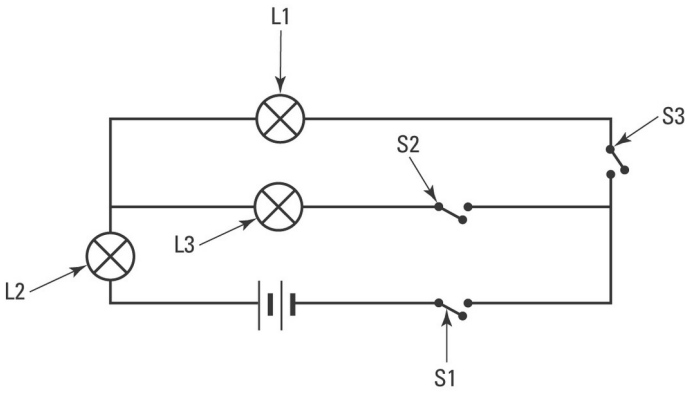
(c) Compare the brightness of the lamps in Circuit 2 to their brightness in Circuit 1. Explain any differences.

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Circuit three

3. Consider the following circuit consisting of 3 lamps and 3 switches.



Predict which lamp(s) will light when:

(a) switches S1 and S2 are closed

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(b) switches S1 and S3 are closed

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(c) switches S2 and S3 are closed.

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